Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	24 September 2026
Team ID	
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:
1)User Interface: Employees submit network requests through ServiceNow catalog items or request forms.

2)Request Processing: Requests are categorized and processed using configured business rules, assignment rules, and Flow Designer workflows.

3)Approval Management: Requests requiring authorization are routed to the appropriate approver.

4)SLA and Notifications: SLA configurations monitor response and resolution targets. Notifications communicate request progress and outcomes.

5)Data and Reporting: ServiceNow tables store request and approval information, while reports and dashboards provide visibility into request status and performance.

Technical Architecture – Automated Network Request Management in ServiceNow

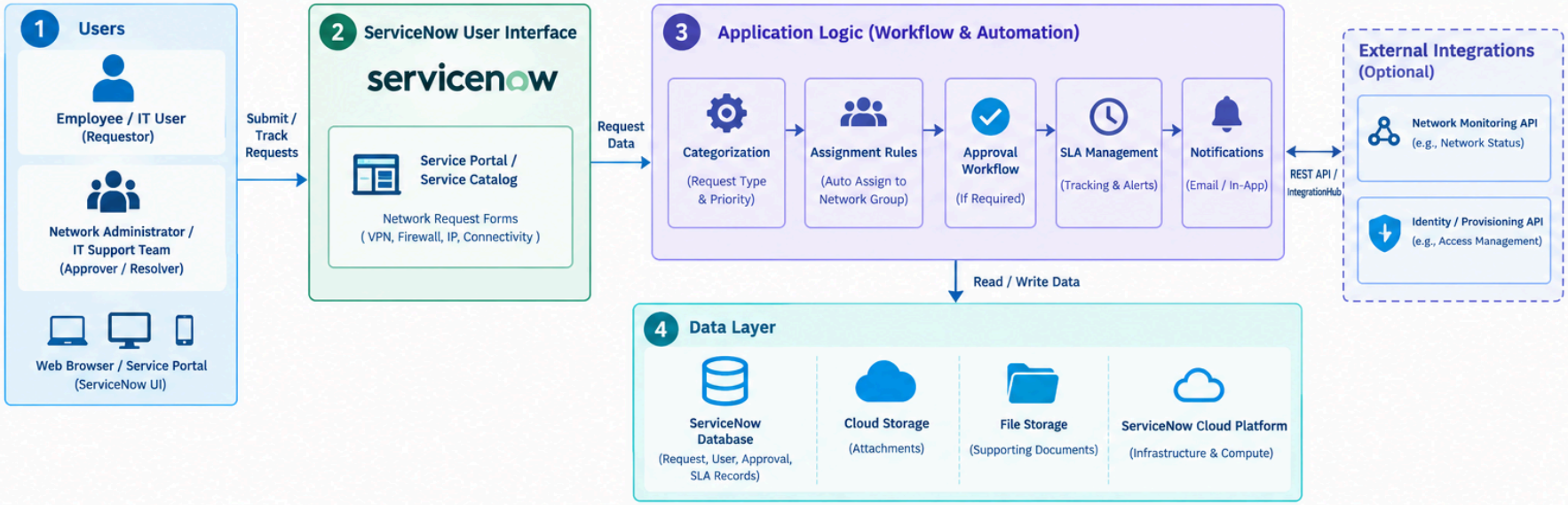


Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How user interacts with application	ServiceNow Service Portal, HTML, CSS, JavaScript
2	Application Logic-1	Request management logic	ServiceNow tables, forms, Business Rules
3	Application Logic-2	Automated assignment logic	Assignment Rules, Flow Designer
4	Application Logic-3	Approval & SLA logic	Flow Designer, Approval Workflows, SLA Definitions
5	Database	Stores request and user data	ServiceNow platform database
6	Cloud Database	Managed cloud data storage	ServiceNow cloud infrastructure
7	File Storage	Stores attachments	ServiceNow attachment storage
8	External API-1	Network monitoring / status	REST API (optional)
9	External API-2	Identity / provisioning	REST API / IntegrationHub (optional)
10	Machine Learning Model	Request classification (optional)	Not required (rule-based can be used)
11	Infrastructure (Server / Cloud)	Application deployment	ServiceNow cloud platform

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Not required for basic implementation	ServiceNow platform, JavaScript, HTML/CSS
2	Security Implementations	Role-based access, ACLs, encryption, HTTPS	ServiceNow roles, ACLs, SSL/TLS
3	Scalable Architecture	Supports additional request types and users	ServiceNow cloud platform, Flow Designer, reusable catalog items
4	Availability	Managed infrastructure and platform availability	ServiceNow cloud platform
5	Performance	Optimized workflows, SLA tracking, reduced manual steps	Business Rules, Flow Designer, SLA Definitions, Reports/Dashboards

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Provides an interface for users to submit, view, and track network requests.	ServiceNow Service Catalog, Service Portal / configurable forms, HTML, CSS, JavaScript
2.	Application Logic-1 Request Management	Creates and manages network requests, including VPN, firewall, IP allocation, and connectivity requests.	ServiceNow tables, forms, catalog items, Business Rules
3.	Application Logic-2 Automated Assignment	Categorizes requests and assigns them to the appropriate network support group based on configured conditions.	ServiceNow Assignment Rules, Business Rules, Flow Designer
4.	Application Logic-3 Approval & SLA Management	Routes requests for approval when required and monitors response and resolution targets.	ServiceNow Flow Designer, Approval Workflows, SLA Definitions
5.	Database	Stores request details, user information, assignment details, approvals, and status records.	ServiceNow platform database and standard tables, such as task/request-related tables .

6.	Cloud Database	Provides managed data storage as part of the ServiceNow platform. A separate cloud database is not required for the basic project.	ServiceNow-managed database infrastructure .
7.	File Storage	Stores request attachments, such as supporting documents or screenshots, when needed.	ServiceNow attachment storage
8.	External API-1	Can integrate with network monitoring or infrastructure tools to retrieve network status, if required.	REST API / ServiceNow REST integration capabilities; optional .
9.	External API-2	Can integrate with external identity or network provisioning services, if required.	REST API / IntegrationHub, depending on licensing and configuration; optional
10.	Machine Learning Model	AI-based request classification is an optional enhancement; the basic project can use rule-based categorization instead.	ServiceNow Business Rules and Flow Designer for rule-based processing; ML not required
11.	Infrastructure (Server / Cloud)	Hosts the application, workflows, database services, and user access.	ServiceNow cloud platform; browser-based client access

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The project primarily uses ServiceNow's built-in platform capabilities. External open-source frameworks are not required for the basic implementation.	ServiceNow platform tools, JavaScript for supported custom scripting, HTML/CSS where applicable
2.	Security Implementations	Role-based access control restricts access to requests and functions according to user roles. Access controls help protect request records	ServiceNow roles, Access Control Lists (ACLs), authentication, HTTPS/TLS,

		and sensitive network information. Approval workflows provide authorization where required.	approval rules
3.	Scalable Architecture	The application uses ServiceNow's managed platform architecture. Request forms, workflow automation, assignment rules, and reporting are configured within the platform, allowing the solution to support additional request types as requirements grow.	ServiceNow cloud platform, reusable catalog items, Flow Designer, configurable tables
4.	Availability	The application is hosted on the ServiceNow platform, which provides managed infrastructure and platform availability according to the applicable service commitments.	ServiceNow cloud infrastructure and platform-managed availability

S.No	**Characteristics**	**Description**	**Technology**
5.	Performance	Request forms, efficient workflow conditions, appropriate assignment rules, and filtered reports help reduce unnecessary processing. SLA monitoring and notifications support timely request handling.	ServiceNow Business Rules, Flow Designer, SLA Definitions, reports and dashboards